



Robot without battery or motor

Researchers have developed an robot that paddles through water using temperature sensitive materials. <https://digit.in/jun18-B33-1>



Wireless RoboFly takes off

The University of Washington have successfully flown the world's first wireless robotic insect. <https://digit.in/jun18-B33-2>



Researchers at the University of Michigan tested the illuminating effects of breast-cancer-detecting pill when exposed to infrared light.

towards being outdated owing to its inability to being able to detect if the cancer is slow-growing or aggressive. Researchers at the University of Michigan have found a way around this old-fashioned system. They are developing a pill, which when swallowed, will illuminate the tumour when exposed to infrared light. The pill can also help identify the nature of the cancer (aggressive or slow) as it reveals information on the types of molecules present in the tumour. The specific technology used to do this is still under development. The research was led by Greg Thurber, a professor of chemical and biomedical engineering. Thurber and his team are now focusing on diversifying the pill's utility by developing methods to make it identify different kinds of tumours as well. His method could do away with the discomfort and inconvenience a mammography entails and could, possibly, entirely replace it to become the primary method of breast cancer detection.

USING ACUPRESSURE TO RELIEVE MENSTRUAL CRAMPS

Cramps, headaches and other discomforts are profusely common ailments faced by women of different age groups during menstruation. The usual ways to treat this, including antibiotics and hormonal contraceptives, can have harsh side-effects. A recent study indicates that



Over 40 percent women state that menstrual cramps have affected their ability to work.

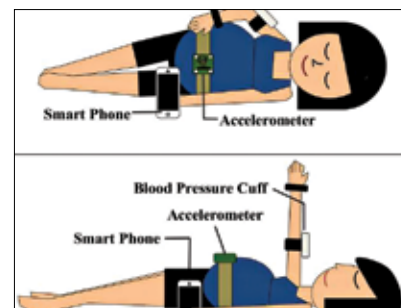
acupressure can largely relieve these symptoms faced by women during their periods. The study showed that an app-guided acupressure treatment, with 221 women participants, showed colossal signs of relief in these participants. Dr. Claudia Witt from Institute of Complementary and Integrative Medicine in Switzerland led the study. She found that the majority of women undergoing these symptoms

are between the ages of 18-35, these women already extensively use their smartphones throughout the day. The app for guided acupressure to relieve periods, called AKUD, would easily fit into their daily schedules and would provide them with the much-needed self-assistance to ease their pain. Most women noted a 50% or higher reduction in the symptoms of menstruation that they faced earlier. Caroline Smith, an acupressure researcher at Western Sydney University in Australia is currently developing an app that integrates the nuances of the study for mass distribution.

APP TO DETECT HEALTH COMPLICATIONS IN PREGNANT WOMEN

Researchers at Purdue University are working towards developing an app and a wearable device that can successfully detect a serious health complication in pregnant women which could lead to harmful conditions in their unborn child or themselves. The condition being sought out

by the app and wearable's sensors is preeclampsia, a complication caused by high blood pressure that can cause grave problems like organ failure or premature birth. Craig Goergen, assistant professor at Purdue's Weldon School of Biomedical Engineering is leading this research. He believes that this invention could radically reduce the number of premature births and prevent further complications in expecting mothers and their children. The researchers have gained much acclaim and have received support and funding from the Bill and Melinda Gates Foundation. WHO has estimated that almost 10% of all maternal deaths in Asia and Africa and 25%



The app and wearable technology detect whether they are susceptible to preeclampsia, a condition that can cause organ damage or early birth.

in Latin America are associated with hypertensive disorders during pregnancy. They believe that these deaths can be easily avoided altogether with prevention and early detection. Currently, the Purdue researchers are on the hunt for companies with technological expertise, who they can partner with to further enhance and perfect the technology.

Technology has geared up to change the face of healthcare and human well-being. Women, especially in the remote areas of the world, require all the assistance that can be dispensed. Technology is a powerful entity that can reach the unseen, the unknown and the needy. With progressive innovations encasing all fragments of human life, technology could very well revolutionise the course of women's healthcare around the world. 